

Muktesh Gawale

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SUMMARY

An MSc Data Science graduate from the University of Nottingham with a strong foundation in Data Analysis, Data Engineering, Machine Learning, AI, and Predictive Analysis. Proficient in Python, SQL, and Power BI, with hands on experience in data profiling, cleansing, integration, and model development for business insights. Certified in AWS Cloud Essentials and Google Analytics, with growing expertise in cloud data services and ETL workflows. Eager to apply my skills and grow as a Data Analyst in a dynamic, data driven environment.

CERTIFICATES

[DataCamp: AI Fundamentals](#) | [DataBricks: Generative AI Fundamentals](#) | [AWS Knowledge: Cloud Essentials](#) | [Google Digital Academy \(Skillsshop\): Google Analytics](#) | [Anaconda Python for Data Science Professional Certificate](#)

EDUCATION

University Of Nottingham

MSc Data Science

Nottingham, United Kingdom

Academic Term: September 2024

Maharashtra Institute of Technology (MIT-WPU) | **GPA: 8.91/10**

BSc Computer Science

Pune, India

Graduation: April 2023

SKILLS

Programming Languages: Python, SQL

Machine Learning Techniques: Supervised Learning, Unsupervised Learning, Regression, Classification

Data Visualization / Preprocessing: Power BI, Matplotlib, Seaborn, Pandas, NumPy

Data Management: Data Profiling, Data Modelling, Data Cleansing, ETL Pipelines, Data Integration

Big Data Tools: Apache Spark

Cloud Platforms: AWS

Statistical Analysis: Regression Analysis, Hypothesis Testing, GLMs

ACADEMIC PROJECTS

Credit Score Classification Using ML & Financial Data

- Conducted comprehensive EDA and data profiling using pandas, seaborn, and plotly, identifying key features impacting credit scores such as annual income, monthly in-hand salary, number of bank accounts, and credit card usage.
- Cleaned and transformed 100K+ financial records, handled <5% missing values, encoded categorical features (e.g., Credit Mix) using LabelEncoder, and normalized inputs to ensure high-quality model readiness.
- Trained and optimised a Random Forest model using scikit-learn, achieving 84.6% accuracy; evaluated classification performance using model accuracy metrics and identified key risk drivers through feature importance analysis.

Predictive Maintenance of Water Pumps in Tanzania

- Executed comprehensive analysis of the "Pump it Up" dataset, encompassing 41 features and 59,400 entries, which enhanced water resource management strategies in Tanzania by identifying critical patterns that led to a 25% reduction in resource wastage.
- Leveraged advanced feature engineering techniques, including the imputation of missing values (under 7%) and Principal Component Analysis (PCA) to preserve 95% variance, enhancing model training efficiency by 30% across diverse algorithms. Utilised and optimised machine learning models including Random Forest, SVM, KNN, and Logistic Regression, achieving a peak prediction accuracy of 79.92% with the Random Forest model outperforming competing models by an average of 6% and enabling more reliable decision making for stakeholders.

Advanced Dementia Prediction Using Big Data and Machine Learning Techniques

- Spearheaded advanced machine learning models for predicting dementia with Lewy bodies, leveraging a comprehensive dataset of 535 samples across 300,000 features, which increased prediction accuracy by 15% compared to previous methods.
- Engineered an innovative data processing pipeline utilising Spark's DataFrame API to enhance data integrity and model input quality, implemented OneHotEncoding and String Indexer techniques that improved feature extraction efficiency by over 30%.
- Executed rigorous model evaluation protocols incorporating key metrics such as accuracy, precision, recall, and AUC, achieved a peak accuracy rate of 68.13% with the Decision Tree Classifier outperforming baseline models by nearly 10%.